

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 12

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Topic

Supplement: Hilbert Variational Gravitation

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Supplement: Hilbert Variational Gravitation. State all assumptions and conventions.

$$S_H = \int d^4x \sqrt{-g} R$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$G_{\mu\nu} = \kappa T_{\mu\nu}$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$N_{pass} = 3/8$$

Show enough intermediate work that another student can reproduce the calculation.